Computational Science at Wabash College

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Department of Mathematics & Computer Science

Ides of August
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Computer Science Minor

Computer Science Minor

Courses Offered

CSC 111	Introduction to Computer Science (CS1)
CSC 112	Advanced Programming (CS2)
CSC 211	Introduction to Data Structures (CS3)
CSC 311	Introduction to Machine Organization (CO1)
CSC 321	Programming Languages (CO4)
CSC 331	Analysis of Algorithms (CO2)
CSC 341	Introduction to Automata, Computability, and Formal Languages (CO3)

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Required for minor:

- Five computer science courses, including CSC 111, 112
- Discrete mathematics course (MAT 108 or 219)

CSC 101 Introduction to Computer Science

Not a programming course

- CS0
- CSC 111 Introduction to Programming

Introduction to the field of computer science

- Data storage and manipulation
- Operating systems
- Networking
- Algorithms
- Programming languages
- Artificial Intelligence
- Theory of computation

Computational Mathematics Track

Requirements

- Calculus I (111), Calculus II (112), Linear Algebra (223), and Abstract Algebra (331)
- Differential Equations (224)
- Numerical Methods (337) or Topics in Computational Mathematics (338)
- One additional course from 219, 226, 314, 337, and 338
- Electives to reach nine credit minimum

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Pure Track

- Calculus I (111), Calculus II (112), Linear Algebra (223), and Abstract Algebra (331)
- Real Analysis (333) or Topology (341)
- Electives to reach nine credit minimum

My Project – Program Review

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Surveys

- Computer science alumni
- Computer science and mathematics students
- Wabash faculty
- Computer science educators (ACM SIGCSE)

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- Consortium for Computer Science in Colleges (CCSC)

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Review curricular models

- IEEE/ACM
- Liberal Arts Computer Science Consortium (LACS)
- Other institutions

Curricular Model Changes

Language of CS1

- Java
- C++
- Scheme
- Python

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Shift introductory sequence

- Introduction to Programming
- Introduction to Data Structures
- Advanced Programming (Second language?)

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Combining and redistributing content of core courses

CO1, CO2, CO3, CO4

Alumni said...

Received good foundation

- Good general overview and theoretical foundation
- Emphasis on concepts, algorithms, logic, problem solving
- Ability think about problems and master tools at disposal
- Advantage over peers in graduate school and careers

Suggest more applications and courses (many applied)

- Artificial intelligence
- Databases
- Networks
- Parallel computing
- Electronics
- Physical devices
- Communications technologies
- Practical business applications

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Suggest a computer science major

Students said...

Strong interest in computational track

- Long list of requirements
- CSC 111 hidden

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Want more computer science and computational courses

- Robotics
- Modeling
- Web design
- Data security

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Want a computer science major

Do not see computational mathematics as a substitute

Faculty said...

Many say computer science minor would be useful to their students

Other courses:

- Informatics
- Database management
- Modeling
- Simulation
- Hardware interfacing
- Interface with economics

Other languages:

- Perl
- Visual Basic
- R, Stata
- Javascript, Actionscript

Faculty said...

Many project opportunities

- Database management
- Econometrics
- Collaborations with art and psychology students
- Historical simulations
- Cognitive science
- Linguistics
- Neural networks
- Hardware control GUI programming

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Small liberal arts institutions

- Fewer than 2500 students
- More in Dept. of C.S. than in combined Dept. of Math. & C.S.
- Almost all offer a major
- More applied than theoretical
- Many offer CS0, but unrelated to the program
- Follow IEEE/ACM guidelines to varying degrees

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Computational Mathematics

May cause department to move apart

Recommendations – Computational Mathematics Track

Proposed changes

- Explicitly list CSC 111
- Eliminate requirement for MAT 224
- Eliminate requirement for additional course from 219, 226, 314, 337, and 338

Proposed Requirements

- Calculus I (111), Calculus II (112), Linear Algebra (223), and Abstract Algebra (331)
- Introduction to Programming (CSC 111)
- Numerical Methods (337) or Topics in Computational Mathematics (338)
- Electives to reach minimum of nine credits in mathematics

Recommendations – Computer Science Minor

Proposed changes

- Eliminate CSC 112
- Create CSC 121 Intro. to Additional Programming Languages
- Retain current CSC 3X1 courses
- Develop new 300-level courses through CSC 271

Proposed Requirements for Minor:

- Five CS courses, incl. CSC 111, 211, and at least half-credit of 121
- MAT 108 or 219

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Long-term

- Expand our offerings
- Develop relationships with other programs
- Develop a computer science major